

MATH 20250 HW6 HINTS AND NOTES

RUOHAN HU

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1. Q5.1.6

The question asks to prove $|(x, y)| = \|x\|\|y\|$ if and only if one of the vectors is a scalar multiple of the other. More than one students proved only the “if” direction.

For the only if direction, a sample proof provided by a student is reconstructed as follows: first of all, what we seek is a solution $t \in \mathbb{C}$ to the equation

$$0 = \|x - ty\|^2 = (x - ty, x - ty) = \|x\|^2 - t(y, x) - \bar{t}(x, y) + |t|^2\|y\|^2.$$

Now, we have

$$|(x, y)|^2 = \|x\|^2\|y\|^2.$$

Then, we have (suppose $y \neq 0$)

$$\begin{aligned} 0 &= \|x\|^2 - \frac{|(x, y)|^2}{\|y\|^2} \\ &= \|x\|^2 - \frac{(x, y)\overline{(x, y)}}{\|y\|^2} - \frac{(x, y)\overline{(x, y)}}{\|y\|^2} + \frac{(x, y)\overline{(x, y)}}{\|y\|^2} \end{aligned}$$

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Let $t = \frac{(x,y)}{\|y\|^2}$. Then, we have $|t|^2\|t\|^2 = t\bar{t}\|y\|^2 = \frac{(x,y)\overline{(x,y)}}{\|y\|^2}$, so the above equation equals to

$$\|x\|^2 - t(y, x) - \bar{t}(x, y) + |t|^2\|y\|^2 = \|x - ty\|^2 = 0.$$

Thus, $x = ty$.

2. Q5.1.4 AND Q5.1.7

Be aware that in these problems we are working in the inner product vector space V in general. The inner product

$$(x, y) = \overline{x_1}y_1 + \dots + \overline{x_n}y_n$$

is only one kind of inner product, on one specific vector space that is $\mathbb{C}^n$ (or $\mathbb{R}^n$).

If a student proved the parallelogram identity based on the inner product above, then such proof is insufficient in generality.

The Parallelogram identity can be proved using only the basic properties (axioms) of inner product spaces in subsection 1.3 of section 5.1 of LADW.

Also, not every vector space is isomorphic to a certain $\mathbb{C}^n$ (or $\mathbb{R}^n$). This is because there are infinite dimensional vector spaces.

3. Q5.2.1

The question asks for vectors that are orthogonal to both of the two vectors $(1, 1, 1, 1)^T, (1, 2, 3, 4)^T$. Thus, the question is asking for the space of solutions to the following system.

$$\begin{cases} x_1 + x_2 + x_3 + x_4 = 0 \\ x_1 + 2x_2 + 3x_3 + 4x_4 = 0 \end{cases}$$

Thus, the answer would be one single vector space of dimension 2. One way of answering it would be to write this space in terms of its two basis vectors.

4. Q5.2.2

$$(\text{Ran}(A^T))^\perp = \ker(A); (\text{Ran}(A))^\perp = \ker(A^T).$$

5. Q5.2.5

The answer is $\ker(A^T)$.

6. Q5.3.5

The answer is $[\frac{7}{6}, \frac{7}{6}, \frac{5}{6}, \frac{1}{2}]$.

7. Q5.3.11

One way of doing the problem. Select an orthonormal basis for E , $v_1, \dots, v_k$, and extend the basis to an orthonormal basis for V , $v_1, \dots, v_n$. Let $Q = [v_1, \dots, v_n]$. Then, we have

$$P = QDQ^{-1}$$

where D is the following diagonal matrix (written as a block matrix)

$$\begin{bmatrix} I_k & 0 \\ 0 & 0 \end{bmatrix}.$$

We obviously have $D^2 = D$ and $D^* = D$, so we easily have $P^* = P$ and $P^2 = P$.

8. A CURIOUS DEDUCTION

Some students believe that

$$\forall v, (v, Av) = 0 \implies A = 0$$

and use this statement in their proofs (of Q5.3.11 or Q5.3.13) without justification. I cannot find such a theorem in LADW.

The statement is, indeed, true in general, if we are working in complex vector spaces, meaning that v can take complex values. (Notice that this would not be true if we are working in real vector spaces.)

The proof, however, is not very easy. We first see that the statement is obvious when A is diagonalizable: in this case we take v to be its eigenvectors, and we see that every eigenvalue of A is 0, so A is similar to the zero matrix, so $A = 0$.

It is a bit more complicated for non-diagonalizable case. Here we use the Upper triangular representation. Suppose under basis $v_1, \dots, v_n$, A is similar to an upper-triangular matrix T . Then, let the entries of T be denoted $T_{i,j}$. Thus, for $v = \sum_{i=1}^n a_i v_i$, we have

$$(v, Av) = \sum_{i,j} \bar{a}_i T_{i,j} a_j = 0$$

Suppose $v = v_i$. Then,

$$T_{i,i} = \bar{a}_i T_{i,i} a_i = (v_i, Av_i) = 0.$$

Thus $T_{i,i} = 0$ for $i = 1, \dots, n$. Suppose $v = v_i + v_j$ and $i < j$. Then $T_{j,i} = 0$ since T is upper-triangular. Thus,

$$T_{i,j} = T_{i,i} + T_{i,j} + T_{j,i} + T_{j,j} = (v_i + v_j, A(v_i + v_j)) = 0.$$

Thus, $T = 0$, so A is similar to the zero matrix, so $A = 0$.

9. Q5.3.12

For the first part $E \subseteq (E^\perp)^\perp$, the key is that for $v \in E$, you need to prove that v is orthogonal to all $w \in E^\perp$. This is true because every $w \in E^\perp$ is orthogonal to every vector in E , v included, so every $w \in E^\perp$ is orthogonal to v , so $v \in (E^\perp)^\perp$.

For the second part $(E^\perp)^\perp \subseteq E$, notice that $V = E \oplus E^\perp$. Thus, for $v \in (E^\perp)^\perp$, we have $v = v_1 + v_2$, where $v_1 \in E \subseteq (E^\perp)^\perp$, so $v_2 = v - v_1 \in (E^\perp)^\perp$. Because v_2 is also in $E^\perp$, we have $v_2 \perp v_2$ i.e. $(v_2, v_2) = 0$, so $v_2 = 0$. Thus, $v = v_1 \in E$.

10. Q6.2.1

The answer is *TF TTTT FTF*.

(a) is true by definition; (b) is obviously false (there are lots of invertible matrices that are not unitary); (d) can be proved using the fact that $(A + B)^* = A^* + B^*$; (g) is false because for example every eigenvalue of $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ is 1, but it is neither orthogonal or unitary; (i) is false by Theorem 5.6.1 of LADW.

For (d) one might read somewhere on stack-overflow that sum of self-adjoint operators might not be self-adjoint, but that is said with respect to functional analysis. In functional analysis, the domain of a self-adjoint operator might not be the whole vector space. Thus those self-adjoint operators are essentially different from what we are taking about here.

11. Q6.2.4

The matrix in the question diagonalizes into

$$\begin{bmatrix} 1 & 0 \\ 0 & 5 \end{bmatrix}$$

The question thus reduced to asking the square root of the above matrix. Thus, the square root is

$$\begin{bmatrix} \pm 1 & 0 \\ 0 & \pm\sqrt{5} \end{bmatrix}$$

The signs of the two entries can be different from each other.

12. Q6.2.5

The statement is false. Notice that the self-adjoint matrices are matrices that are equal to its conjugate transpose. For example,

$$A = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$$

is not self-adjoint, because its conjugate transpose is

$$A^* = \begin{bmatrix} 1 & 0 \\ 0 & -i \end{bmatrix}.$$

On the other hand, notice that

$$A^2 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

is indeed a self-adjoint matrix. Thus, the square root of a self-adjoint matrix is not necessarily self adjoint.

13. Q6.2.7

The answer is FT .

14. Q6.2.8

Hint: use Theorem 6.2.2 of LADW: if a matrix is self-adjoint then it is diagonalizable (by a unitary matrix). Also, notice that sum of self-adjoint matrices are also self-adjoint.

15. Q6.2.14

Hint: use Theorem 6.2.4 of LADW: if a matrix is normal, then it is diagonalizable (by a unitary matrix).

16. SUPPLEMENTARY 1

Indeed every non-zero vector v in $(\mathbb{Z}/5\mathbb{Z})^2$ has exactly 5 orthogonal vectors. The easiest proof I can come up with is as follows.

We first notice that the vectors orthogonal to v forms a vector space $V^\perp$. $V^\perp$ cannot be two dimensional, because otherwise v would be orthogonal to every vector in $(\mathbb{Z}/5\mathbb{Z})^2$, leading to $v = 0$ contradiction. $V^\perp$ must contain at least one vector other than 0, by some argument. Thus, we must have $\dim(V^\perp) = 1$, and a one-dimensional vector space over $\mathbb{Z}/5\mathbb{Z}$ must have 5 elements. Thus, there are exactly 5 vectors orthogonal to v .

17. SUPPLEMENTARY 2

- (i) Sometimes; (ii) Always; (iii) Sometimes.
- (i): a vector with norm not 1 can never be part of an orthonormal basis.

18. SUPPLEMENTARY 3

- (i) Never; (ii) Sometimes; (iii) Sometimes.